

Academic Review Report R3

Leverage Direction Matters: Cross-Asset Evidence on GARCH Model Selection and Volatility Targeting

Author: Yi-Hao Lai

Document Version: main_v2.tex + body_v2.tex + tables.tex (April 2026)

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Review Standard: Strict (JBF Referee Standard)

Reference: R2 review dated 2026-04-05; K899 unified VaR; K902 Tables 1 & 3 supplement

R3 Scorecard

R2 Issue	R3 Status	Severity
C1: HM γ contradiction	RESOLVED (R2)	—
C2: Kupiec p -values falsely equalized	RESOLVED	—
C3: Table 5 cherry-picks from multiple expts	PARTIALLY RESOLVED	MAJOR
C4: Table 1 descriptive stats untraceable	RESOLVED (with caveats)	MINOR
C5: Table 3 QLIKE untraceable	NOT RESOLVED — scale mismatch	SEVERE
N1: K899 diverges from Table 5	ACKNOWLEDGED, not acted on	MAJOR
N2: K899 ES results not incorporated	UNCHANGED	MAJOR
N3 (NEW): QLIKE scale mismatch K902 vs paper	NEW	SEVERE

SEVERE count: 1 (down from 4 in R2).

Overall assessment: Minor–Major Revision. The core conceptual issues are solved. One severe traceability gap remains (Table 3 QLIKE scale) plus three major items that would strengthen the paper substantially.

Detailed Assessment of Each Issue

C2: Kupiec p -values

RESOLVED

R2 flagged that Table 5 (tab: var_ortho) reported $p = 0.60$ for both GJR+Student- t and GJR+HistSim. The current tables.tex (lines 95–96) now shows:

- GJR+Student- t : Kupiec $p = 0.67$
- GJR+HistSim: Kupiec $p = 0.64$

These are the correct values from K802/K824v2 for the 2023–2024 window ($N = 502$). No further action needed.

C3: Table 5 Cherry-Picking

PARTIALLY RESOLVED

The current tables.tex (lines 100–103) adds a note referencing K899’s extended OOS (2020–2025, $N = 1,508$). This is a constructive step: the reader now knows that a unified experiment exists and can judge the sensitivity to the evaluation window. However:

1. The **main table values still come from the original mixed-source** (K799/K802/K824v2 for the 2023–2024 window). The four rows are now at least *internally consistent* (same 502-day period,

same Kupiec correction), but the underlying estimation settings (refit frequency, expanding vs. rolling window) may still differ across the original experiments.

2. K899 data is mentioned only in a footnote, not incorporated as a robustness table.
3. The note says “all methods fail at 5% due to COVID/2022 turbulence”—this is factually imprecise. At the 5% VaR level under K899, most methods *pass* Kupiec (e.g., GJR+HistSim Kupiec $p = 0.59$). The failures are in Basel traffic light (Red for all methods at 5%).

Residual severity: MAJOR. The 2023–2024 values are no longer falsely equalized (C2 fixed), but the single-source requirement is not fully met. Recommend either: (a) re-extract the 2023–2024 subset from K899 to guarantee identical settings, or (b) add K899 as a formal robustness table in the appendix.

C4: Table 1 Descriptive Statistics

RESOLVED (with caveats)

K902 now provides complete descriptive statistics for 9 assets with traceable methodology. Verification against `tables.tex`:

Asset	Stat	Paper	K902	Match?
SPY	Mean (%)	0.063	0.062	Off by 0.001
SPY	Std (%)	1.16	1.16	Yes
SPY	Skewness	−0.32	−0.32	Yes
SPY	Kurtosis	14.6	14.6	Yes
SPY	N	2260	2262	Off by 2
QQQ	Std (%)	1.45	1.44	Off by 0.01
EEM	Mean (%)	0.036	0.037	Off by 0.001
BTC	Mean (%)	0.202	0.203	Off by 0.001
BTC	N	3285	3287	Off by 2

Assessment: The discrepancies are tiny—within rounding and likely due to (a) 2 additional trading days as data was re-downloaded, and (b) different rounding conventions (0.0625% rounds to either 0.062% or 0.063% depending on precision). These are **not substantive errors**. The paper should either: (i) update Table 1 to exactly match K902, or (ii) note that K902 was run on a slightly later data pull. The N discrepancy (± 2 observations) should be resolved by fixing the exact end date.

Residual severity: MINOR. K902 provides the traceable source. The rounding-level discrepancies do not affect any conclusion.

C5 + N3: Table 3 QLIKE — Scale Mismatch

SEVERE

This is the single remaining severe issue. K902 provides cross-asset QLIKE for 6 assets $\times$ 2 periods, but the values do not match the paper’s Table 3:

Cell	Paper	K902	Difference
SPY 2023–24 GARCH	−8.985	−8.632	−0.353
SPY 2023–24 GJR	−9.034	−8.681	−0.353
SPY 2025 GARCH	−8.719	−8.321	−0.398
SPY 2025 GJR	−8.818	−8.410	−0.408
QQQ 2023–24 GARCH	−8.554	−7.958	−0.596
QQQ 2023–24 GJR	−8.475	−7.968	−0.507
GLD 2023–24 GARCH	−9.058	−8.432	−0.626
TLT 2023–24 GARCH	−9.169	−8.156	−1.013
BTC 2023–24 GARCH	−6.871	−6.319	−0.552

Diagnosis: The offsets are **not constant**—they range from -0.35 (SPY) to -1.01 (TLT). This rules out a simple additive constant difference. Two likely causes:

1. **Different QLIKE formula conventions.** The paper’s Eq. (4) defines $QLIKE = \frac{1}{T} \sum (\frac{r_t^2}{h_t} + \ln h_t)$. Some implementations use $\ln(h_t) + r_t^2/h_t - 1 - \ln(r_t^2)$, which adds the term $-1 - \ln(r_t^2)$. Since $\ln(r_t^2)$ varies across assets (more negative for low-vol assets like TLT), this would explain both the sign and the asset-dependent offset.
2. **Different estimation windows.** If K902 uses a different rolling window size (w) or refit frequency than the original experiment that produced the paper’s numbers, the GARCH parameter estimates—and hence the forecast sequence h_t —will differ, producing different QLIKE means.

Either way, the paper’s Table 3 values are not traceable to K902. The original source experiment for Table 3 is unknown (K902 was supposed to *replace* it), but K902 produces systematically different numbers.

Required action:

1. Determine which QLIKE formula the paper’s Table 3 originally used (check the original experiment script, if it exists).
2. If K902 uses the correct Patton (2011) proxy-robust form, **update Table 3 to K902’s values.**
3. If the paper’s values are correct and K902 has a formula bug, fix K902.
4. Either way, the **DM test statistics and p -values must come from the same experiment as the QLIKE values.** Currently the paper reports DM $p = 0.001$ for SPY, while K902 reports DM $t = 4.221$, $p = 0.000$. The qualitative conclusion (GJR wins for SPY) is unchanged, but the specific numbers differ.
5. The `body_v2.tex` text at line 347 quotes “ $QLIKE \approx -9.034$ ” for SPY’s GJR—this must match whichever source is adopted.

Additionally, K902 reveals a noteworthy finding: **QQQ 2023–24 reverses sign.** The paper reports $\Delta = +0.92\%$ (GARCH wins, $p = 0.067$), while K902 reports $\Delta = -0.12\%$ (GJR slightly wins, $p = 0.619$, NS). This does not affect the paper’s narrative (both are non-significant for 2023–24, and GJR wins significantly in 2025), but the sign reversal in the Δ column would change one cell in Table 3.

Severity: SEVERE. This is the paper’s most important empirical table. Untraceable QLIKE values undermine the reproducibility claim.

N1 + N2: K899 Integration

MAJOR (combined)

K899 provides:

- Unified VaR backtest (7 methods, 1508 OOS days, identical settings)
- ES backtest (Acerbi-Szekely): only GJR+HistSim passes ($z = 1.64$, $p = 0.10$)
- Fissler-Ziegel joint VaR-ES scores: GJR+HistSim best at both 1% and 5%

The current paper uses K899 only in a footnote (lines 100–103 of `tables.tex`). The ES results—arguably the paper’s strongest practical finding and directly relevant to Basel III (which now requires ES over VaR)—are not mentioned at all.

Recommendation:

1. Add an appendix table with K899’s full 7-method results (VaR + ES columns).
2. Add 1–2 sentences to Section 4.6 or the conclusion noting that GJR+HistSim is the only method passing both VaR and ES tests. This strengthens the orthogonality narrative and the complexity ceiling argument without requiring a major rewrite.
3. Correct the footnote: at 5% VaR, methods pass Kupiec but fail Basel traffic light due to clustering—not “all methods fail.”

R2 Major Issues — Quick Update

ID	Issue	R3 Status	Note
M1	Paper length (62 pp)	UNCHANGED	Not addressed; lower priority
M2	Abstract overclaims	RESOLVED (R2)	—
M3	Proposition 1 mechanical	IMPROVED (R2)	—
M4	Harvey $t > 3.0$ inconsistency	UNCHANGED	K902 now shows SPY DM $t = 4.22$ (Harvey pass), but body text still quotes $p = 0.001$ from old experiment
M5	No ES analysis	DATA READY	K899 has ES data; not yet in paper
M6	DeMiguel mischaracterized	UNCHANGED	Line 46: “hybrid implied-realized” should be “conditional multifactor”
M7	Hybrid VT Sharpe rounding	UNCHANGED	
M8	Missing references	UNCHANGED	Bekaert & Wu (2000), Figlewski & Wang (2000)

Summary and Action Plan

SEVERE (must fix): 1 item

1. **Resolve Table 3 QLIKE scale mismatch.** Determine which formula (paper or K902) is the Patton (2011) standard. Update Table 3 + body text to match the traceable experiment. Update DM statistics accordingly. *[C5 + N3]*

MAJOR (strongly recommended): 3 items

2. **Unify Table 5 source.** Either re-extract 2023–2024 from K899 or add K899 as appendix table. Fix footnote wording. *[C3]*
3. **Incorporate K899 ES results.** Add ES pass/fail column or appendix table. Mention in text. *[N2, M5]*
4. **Reconcile Table 1 rounding.** Update 5 cells to match K902 exactly (N , means). *[C4 residual]*

MINOR (desirable): 5 items

5. Fix DeMiguel characterization (M6, line 46).
6. Add Bekaert & Wu (2000) and Figlewski & Wang (2000) references (M8).
7. Standardize Harvey threshold usage (M4).
8. Report Hybrid VT Sharpe as 0.985 (M7).
9. Move secondary analyses to supplementary material (M1).

Bottom line: The paper has improved substantially from R2 (4 SEVERE $\rightarrow$ 1 SEVERE). The conceptual issues are all resolved. The single remaining severe item—Table 3 QLIKE traceability—is a mechanical fix once the formula convention is determined. The three major items (Table 5 unification, ES incorporation, Table 1 rounding) would strengthen the paper but are not blocking for a first submission if addressed in cover letter promises. With these fixes, the paper reaches a submittable state for JBF.

Reviewer: Claude Opus 4.6 (1M context), acting as JBF referee. R3 review, 2026-04-05.

Sources verified: K899 (experiments/k899_unified_var_paper1_results.json), K902 (experiments/k902_tables.tex, body_v2.tex).